

Central Nervous System (CNS) Pharmacology

For HO & Nursing Stud. 2nd Yr

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Drugs Acting on CNS

Introduction

- CNS drugs act on brain and spinal cord
 - Medical uses (treatment of mental illnesses, suppression of seizures, relief of pain, and production of anesthesia) and
 - Nonmedical uses (stimulants, depressants, euphorants, and other “mind-altering” uses).
- ***Transmitters of the CNS***
 - More than a dozen neurotransmitters
 - Physiological functions of the main CNS neurotransmitters are discussed briefly.

➤ **Inhibitory neurotransmitter (aa)**

- Gamma amino butyric acid (GABA)
- Glycine

➤ **Excitatory neurotransmitter (aa)**

- Glutamate
- Aspartate

❖ Gamma aminobutyric acid (GABA)

- Inhibitory amino acid
 - ✓ Prevent a nerve fiber from excitation
- Reduced function/amount of GABA lead to excessive stimulation that results in *seizure*
- GABA action is mediated by two types of receptors:
 - ✓ GABA_A receptors (ligand-gated ion channels) and
 - Selectively permeable to Cl⁻
 - ✓ GABA_B receptors (G-protein coupled)
 - Either inhibit Ca²⁺ channels or activate K⁺ channels

❖ Glycine

- Inhibitory amino acid
- Mainly limited to the spinal cord
- Glycine receptor is selectively permeable to Cl^-
- Blockade of glycine receptors cause seizure

❖ Glutamate

- **Excitatory** neurotransmitter
- Act at different receptors:
 - ✓ NMDA, AMPA, Kainate and metabotropic glutamate receptors
- NMDA receptor is a potential target for **ketamine**
- Excessive stimulation may lead to **Seizure**

- *Noradrenaline (Norepinephrine):*
 - Noradrenergic transmission is important in
 - » control of mood (functional deficiency resulting **depression**)
 - » wakefulness, and
 - » alertness.

- ***Dopamine:***
 - Dopamine is important in
 - Motor control,
 - Has behavioral effects (excessive dopamine activity is implicated in **schizophrenia**)
 - Important in hormone release (prolactin, GH)
 - Chemoreceptor trigger zone causes nausea and vomiting.

- ***Serotonin (5-HT):***

- Physiological functions associated with 5-HT pathways include;

- ✓ Feeding behavior

- ✓ Behavioral response (hallucinatory behavior)

- ✓ Control of mood and emotion,

- ✓ Control of body temperature

- ✓ Vomiting

- *Acetylcholine(Ach)*: has effects
 - ✓ On arousal
 - ✓ Movement
 - ✓ On learning, and short-term memory
 - ✓ Dementia and parkinsonism are associated with abnormalities in cholinergic pathways.

- **Drugs for Parkinson's Disease (PD)**

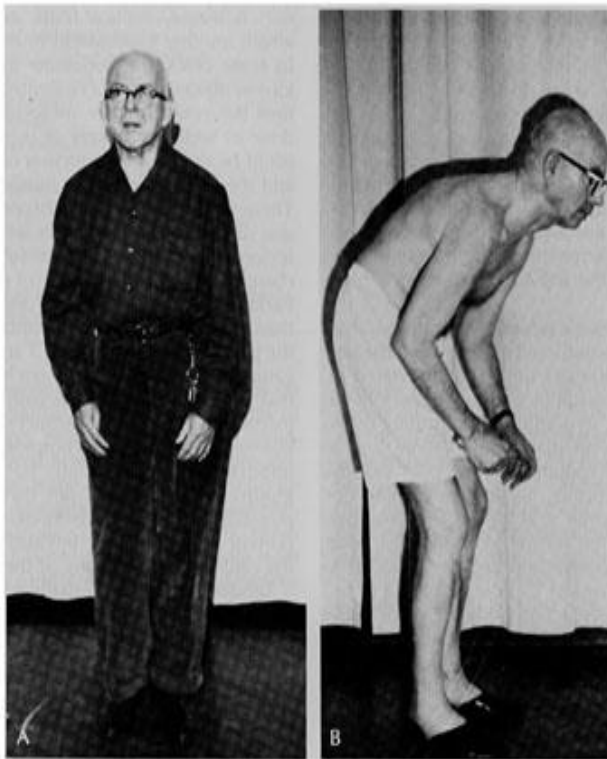
- Parkinson's disease is a neurodegenerative disorder characterized by

- tremor,
 - rigidity,
 - postural instability, and
 - Bradykinesia (slowed movement).

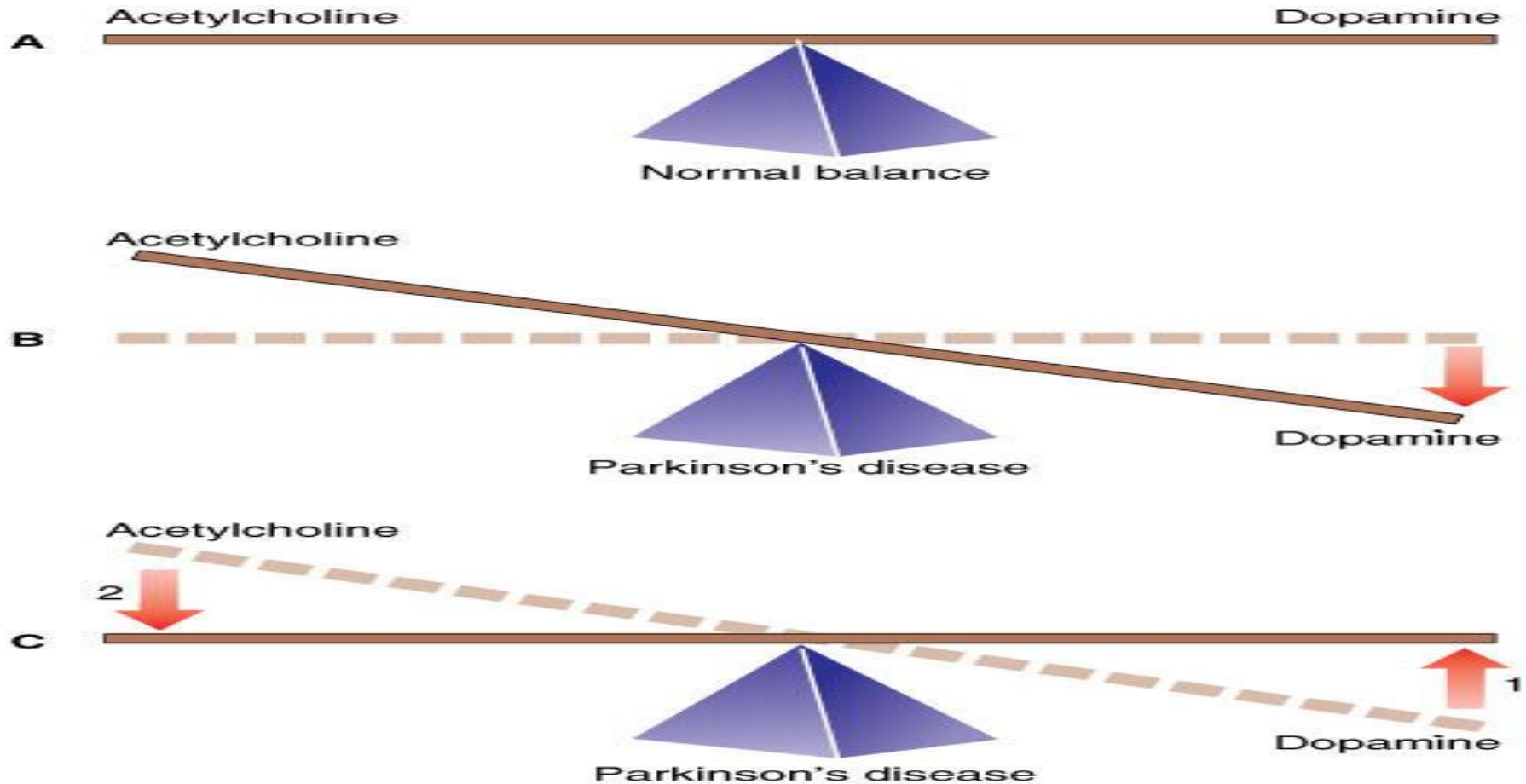
- Cause of Imbalance (**DA & ACh**)—Degeneration of neurons supplying **DA** to **striatum**; therefore, excitatory influence of ACh is unopposed leading to release of GABA

Parkinsonism (PD) - signs

Parkinson's Disease



Parkinson's disease



- A,** Normal balance of acetylcholine and dopamine in the CNS.
B, In Parkinson's disease, a decrease in dopamine results in an imbalance.
C, Drug therapy in Parkinson's disease is aimed at correcting the imbalance between acetylcholine and dopamine. This can be accomplished by either
 1. increasing the supply of dopamine or
 2. blocking or lowering acetylcholine levels.

Fig. 14-1. The neurotransmitter abnormality of Parkinson's disease.

Drug Therapy

A. **Therapeutic Goal**—To improve ability for activities of daily life (only symptomatic relief)

B. **Treatment Strategy**—Restore balance between DA and ACh by **activating** DA receptors or **blocking** ACh receptors

C. **Drugs Employed**—Two major

- categories (dopaminergic drugs and
- anticholinergic drugs).

Drug	Mechanism of Action	Therapeutic Role
DA Precursor Levodopa Levodopa/carbidopa	Levodopa undergoes conversion to DA in the brain and then activates DA receptors (carbidopa just blocks destruction of levodopa in the periphery)	First-line drug, or supplement to a dopamine agonist Why not DA itself????
DA Agonists Bromocriptine Pergolide Pramipexole Ropirinole	Direct activation of DA receptors	First-line drug, or supplement to levodopa (the nonergot agents-pramipexole and ropirinole- are preferred)
COMT Inhibitors Entacapone Tolcapone	Inhibit breakdown of levodopa by Catechol-O-methyl transferase (COMT)	Adjunct to levodopa (to increase “’ontime” and decrease “wearing off”)
DA Releaser Amantadine (antiviral agents)	Promotes release of DA from remaining DA neurons may also block DA reuptake	Second line therapy for motor fluctuations
MAO-B Inhibitor Selegiline Rasagiline	Inhibits the breakdown of DA by monoamine oxidase type B (MAO-B)	Used in newly diagnosed patients for possible neuroprotection; second or third-line agent as adjunct to levodopa

❑ Centrally Acting **Anticholinergic** Drugs:

(Benzhexol/ trihexyphenidyl, benztropine)

- First drugs used was drugs like atropine but caused **intense** anticholinergic effects so now classic ones are obsolete.
- ***Mechanism of action*** help control symptoms by blocking access of ACh to receptors.
- ***Therapeutic use*** **less** effective than levodopa but **less** serious adverse effects; good for **mild** disease in **young** patients; **avoid** in elderly.
- ***Adverse effects*** peripheral and CNS effects
 - » Atropine like side effects; dry mouth, constipation, tachycardia...

Drugs for Epilepsy

Epilepsy :

- Refers to group of disorders characterized by excessive excitability of neurons within the CNS.
- ***Seizure Generation***—initiated by synchronous, high-frequency discharge from group of hyper excitable neurons (focus)
- Symptoms depend on location of focus.
- ***Types of Seizures***—Two Broad Categories

A. Partial (Focal) Seizures

- » 1. Simple partial—No effect on consciousness and may involve single limb
- » 2. Complex partial—Impairment of consciousness, which may be seen as confused or bizarre behaviour.

B. Generalized Seizures—May be convulsive or non convulsive. As a general rule, they produce **immediate loss of consciousness**.

SEIZURES

Partial

(consciousness preserved)

— **Simple (consciousness normal)**

— **Complex (consciousness altered/
no memory)**

Generalized

(consciousness lost/no memory)

— **Tonic-clonic**

— **Absence**

— **Myoclonic**

— **Infantile spasm**

— **Status epilepticus**

Generalized Status Epilepticus

(convulsive or non-convulsive)

Partial Status Epilepticus

(consciousness lost/no memory)

- **How Anti-seizure Drugs Work (Mechanisms)**

- Suppress discharge of neurons within a seizure focus
 - suppress propagation of seizure activity from focus to other brain areas.
- **Suppression of Sodium Influx**—Phenytoin, carbamazepine, valproic acid, and lamotrigine reversibly bind to sodium channels.
- **Suppression of T-type calcium channel** — Valproic acid and ethosuximide used for **absence** seizures
- **Enhancement of GABA:** Benzodiazepines (e.g. Diazepam, clonazepam,...) and Barbiturate (e.g. Phenobarbitone)...used also as sedative-hypnotics
- **Inhibit the excitatory neurotransmission (Glutamate):**
Topiramate and felbamate

Table 7. Drugs for specific types of seizures

Seizure Type	Drugs used for treatment		
	Effective and well tolerated	Effective but less tolerated	Newer Alternatives
Partial Seizures			
Simple Partial	Carbamazepine Phenytoin Valproic acid	Clorazepate Phenobarbitone Primidone	Gabapentin Lamotrigine Topiramate Tiagabine
Complex partial	Same as simple partial	Same as simple partial	Same as simple partial
Generalized seizures			
Tonic-clonic (grand mal)	Carbamazepine Phenytoin Valproic acid	Phenobarbitone Primidone	Lamotrigine Topiramate
Absence (petit mal)	Ethosuximide Valproic acid	Clonazepam	Lamotrigine
Myoclonic	Valproic acid Clonazepam	-	Lamotrigine Topiramate
Atonic	Same as myoclonic	-	Same as myoclonic

Sedative-Hypnotic Drugs

- Anxiety and insomnia are common complaints
- We can use same drugs with different dosages
 - Drugs used for anxiety called “antianxiety agents” or “anxiolytics”
 - Drugs used for sleep known as “hypnotics”
- **Benzodiazepines** — First choice for treating anxiety and insomnia;
 - Most produce identical effects (Diazepam, alprazolam, clonazepam and others)

- ***Molecular Mechanism of Action***—Potentiates the actions of gamma-aminobutyric acid (**GABA**), which is an inhibitory neurotransmitter.
- ***Therapeutic uses***—Three major uses: anxiety, insomnia, and seizure disorders; also for muscle spasm, ethanol withdrawal, panic disorder, and anesthesia induction

- ***Adverse effects***—CNS depression, anterograde amnesia, paradoxical effects, respiratory depression, and low incidence of abuse.
- ***Tolerance and physical dependence***—Little or no tolerance with anxiolytic and hypnotic effects but significant **tolerance** side effects

- **Benzodiazepine-Like Drugs—zolpidem and zaleplon**
 - **A. Zolpidem** —For short-term management of insomnia; binds to GABA receptors
 - Side effects of daytime drowsiness and dizziness.
 - **B. Zaleplon** —New class of hypnotics; rapid onset, short duration; helps for falling asleep but not for staying asleep
- **Barbiturates** — (Phenobarbitone, secobarbital, thiopental)

Nonselective depression of CNS; can be used for daytime sedation, induction of sleep, suppression of seizures, and general anesthesia (powerful **respiratory depressants** that can be fatal with overdose).

Antipsychotic Agents

- **Clinical presentation**

- Schizophrenia is a chronic psychotic illness characterized by
 - ✓ Disordered **thinking** and
 - ✓ Reduced ability to comprehend **reality**.

- **Etiology**

- Although there is strong evidence that schizophrenia has a **biologic basis**, the exact etiology is unknown
- Possible primary defects include
 - ✓ Excessive activation of CNS receptors for dopamine, and
 - ✓ Insufficient activation of CNS receptors for glutamate

Conventional (Typical) Antipsychotic Agents I: Group properties

Mechanism of action

- Suppress symptoms of psychosis by blocking **D2 receptors** in the **mesolimbic** area of the brain
- ***Classification***
 - ✓ Can be classified by potency or chemical structure.
 - ✓ From a clinical view point, classification by **potency** is more helpful.
- *Classification by potency*
 - Conventional antipsychotic agents can be classified as **low potency**, **medium potency**, or **high potency**.

Atypical (2nd generation) antipsychotics

Mechanism of action

Block D2 as well as **5-HT₂** receptors

E.g.

- Clozapine (reserved for **resistant** types of schizophrenia)
- Risperidone
- quetiapine

Antipsychotic Drugs: Relative Potency and Incidence of Side Effects

		Incidence of Side Effects			
Drug	Equivalent oral dose (mg)	Extrapyramidal effects	Sedation	Orthostatic hypotension	Anticholinergic effects
Conventional agents Low potency Chlorpromazine Thioridazine Moderate potency Triflupromazine Perphenazine Loxapine High potency Haloperidol Fluphenazine	 100 100 25 10 10 2 2	 low Low Moderate Moderate Moderate High High	 High High High Moderate Moderate moderate Low Low	 High High Moderate Low Low low Low Low	 Moderate High Moderate Low Low Low Low
Atypical Agents Clozapine Risperidone quetiapine	 50 4	 Very low Very low	 High Low	 Moderate Low	 High None

- ***Therapeutic Uses Antipsychotics—***

- Schizophrenia primary reason for antipsychotic drugs to suppress symptoms (not cure) and help prevent relapse,
- Treatment of bipolar disorder
- Prevention of emesis

Antidepressants:

- Antidepressants are used primarily to relieve symptoms of depression

Clinical Features of Depression:

- The principal symptoms are depressed mood and loss of pleasure or interest in all or nearly all of one's activities and interests

Pathogenesis:

» monoamine hypothesis of the depression

Caused by a functional insufficiency of monoamine NTs (noradrenaline, serotonin or both).

Treatment Modalities:

- Pharmacotherapy
- depression-specific psychotherapy

1. Tricyclic Antidepressants (TCAs)

- ✓ **Amitryptiline**, nortryptiline
- ✓ Imipramine, desipramine
- First-choice drugs
- ***Mechanism of Action***—Block monoamine reuptake (**NA & 5-HT**)
- ***Pharmacokinetics***—Long and variable half-lives; given in single daily dose

Therapeutic Uses:

- a. Depression—takes 2–3 weeks for initial responses
- b. Bipolar disorder, chronic insomnia, attention-deficit hyperactivity disorders (ADHD)

Adverse Effects—Orthostatic hypotension is most common; anticholinergic effects; sedation; **cardiac toxicity is most serious**; seizures; hypomania.

2. Selective Serotonin Reuptake Inhibitors (SSRIs)—Effective as TCA without hypotension, sedation, or anticholinergic effects; overdose **doesn't cause cardiotoxicity**.

e.g. Fluoxetine, sertraline , fluvoxamine , paroxetine , Citalopram

❑ Fluoxetine:

- The most widely prescribed antidepressant since it is **as effective as TCAs**, has **fewer side effects**, and is less dangerous; but should be avoided taking with MAOIs
- ***Mechanism of action***—**Selective inhibition** of serotonin (5-HT) reuptake to intensify transmission at serotonergic synapses; produces **CNS excitation** rather than sedation.
- ***Therapeutic uses***—For major depression, obsessive compulsive disorder (OCD), panic disorder, premenstrual disorder

- **Pharmacokinetics**—Well absorbed regardless of food; 94% protein bound; fluoxetine (half-life 2 days) hepatic conversion to norfluoxetine (half-life 7 days)
 - **Adverse effects**—Sexual dysfunction (70%); nausea (21%); headache (20%); CNS stimulation including nervousness (15%), insomnia (14%), and anxiety (10%); also weight gain up to 20 lb;
Other SSRIs—sertraline , fluvoxamine , paroxetine , Citalopram
- None should be combined with MAOIs
- Because both produces serotonin syndrome

3. Monoamine Oxidase Inhibitors (MAOIs-A) - Phenelzine, Tranylcypromine

- Second- and third-choice drugs especially since major problem is **hypertensive crisis** with tyramine foods.

Opioid (Narcotic) Analgesics

- Is any drug, natural or synthetic, with actions like morphine;
- Narcotic refers to opioids, cocaine, marijuana, and LSD
- enkephalins, endorphins, and dynorphins serve as neurotransmitters

- ***Opioid Receptors***—Mu (the most important), kappa, and delta.
 1. Mu (μ) receptor- activation includes analgesia, respiratory depression, euphoria, miosis and constipation.
 2. Kappa (κ) receptor- activation produces analgesia, sedation, and dysphoria.
 3. Delta (σ) receptors- are responsible for producing spinal/supraspinal analgesia

• *Classification of Drugs that Act at Opioid Receptors*

1. Pure opioid agonists—Activate mu and kappa and divided into strong opioid agonists (morphine) and moderate-to-strong opioid agonists (codeine).
2. Agonist-antagonists opioids—Five are available (pentazocine, nalbuphine, butorphanol, dezocine, and buprenorphine); when administered alone, produce analgesia, but given to a person taking opioids, produce an antagonistic effect.
3. Pure opioid antagonists- **Naloxone, naltrexone**
 - ✓ **Antagonize** mu and kappa receptors and are used for **reversal of respiratory and CNS depression** due to opioid agonists **overdoses**

- **Basic Pharmacology of the Opioids**

Morphine:

- Produces analgesia, sedation, euphoria, respiratory depression, cough suppression, and decreased bowel motility.
- ***Mechanism of analgesic action***—Mimics endogenous opioid peptides (mu and kappa receptors).
- ***Therapeutic use***—Relief of pain (moderate to severe) for labor and delivery, MI pain and dyspnea associated with LVF and pulmonary edema, and preoperatively for sedation and anti-anxiety.

- ***Adverse effects***—Respiratory depression, constipation, orthostatic hypotension, urinary retention, cough suppression, biliary colic caused by morphine, emesis, euphoria/dysphoria, sedation, and miosis.
 - ***Abuse liability***
 - ***Toxicity***—Triad of signs: coma, respiratory depression and pinpoint pupils—provide support and give antagonist (naloxone).
 - **Other Strong Opioid Agonists-pethidine , methadone, levomethadyl, heroin, fentanyl** (can be given parenterally, transdermally, transmucosally, or as lozenges or lozenges on a stick).

- Morphine-dose-10-15mg oral
- **Codeine**-10-30mg to suppress cough (now **banned** in Ethiopia)
- Tramadol-50-100mg oral/im/iv-analgesic
- **Pethidine**-50-100mg, IM, Sc- **preferred** analgesic during **labor**

Pure opioid antagonist:

- **Naloxone** –at 4-10 mg dose it also antagonizes the action of morphine (so used to treat opioid toxicity)
- Is a drug of choice for morphine poisoning
- For **reversing neonatal** asphyxia due to opioid use during labor

Stroke

- Can be ischemic (87%) or hemorrhagic (13%) stroke.
- Transient ischemic attacks (TIAs) require urgent intervention to reduce the risk of stroke.
- Antiplatelet therapy is the cornerstone for 2° prevention of noncardioembolic ischemic stroke.
 - ✓ Drugs like Aspirine, clopedogrel, dipyridamole
- Oral anticoagulation is recommended for the 2° prevention of cardioembolic stroke in patients with atrial fibrillation.
 - ✓ Drugs like warfarin, rivaroxaban, and apixaban
- BP lowering is effective in both the 1° & 2° prevention of both ischemic and hemorrhagic stroke.
 - ✓ Labetalol, sodium nitropruside, nicardipine
- BP lowering in the acute ischemic stroke period (first 7 days) may result in ↓ cerebral blood flow and worsened symptoms.

- ❑ The goals of treatment of acute stroke are:
 - a) To reduce the ongoing neurologic injury and ↓ mortality and long-term disability
 - b) To prevent complications 2° to immobility and neurologic dysfunction
 - c) To prevent stroke recurrence.

- **N.B.** There are currently no standard pharmacologic strategies for R_x of hemorrhagic stroke. Instead recommended to manage the BP, raised ICP, and other medical complications (nimodipine).

- The initial approach to the patient with a presumed acute stroke is
 - ✓ To ensure that the patient is supported from a respiratory and cardiac standpoint
 - ✓ To quickly determine whether the lesion is ischemic or hemorrhagic (CT scan)
- Patients with elevated BP should remain **untreated** unless
 - ✓ Their BP > 220/120 mm Hg, or
 - ✓ They have evidence of aortic dissection, acute MI, pulmonary edema, or hypertensive encephalopathy.
 - ✓ If BP is treated, short-acting parenteral agents: labetalol and nicardipine, are favored.
- tPA and statins has a crucial role in stroke management.

Thanks!